

PAPER_816: EHT, ngEHT, and BHEX New SMBH Population Photon Ring UQFF

Unified Quantum Field Framework — Whitepaper 816

Session: 192 | **Version:** v5.48 | **Date:** April 4, 2026 **Source:** grok_share_0d888ea9-50be.txt (June 13, 2025 04:45 PM); “Accessing_a_New_Population_of_Supermassive_Black_H.pdf” (ApJ May 2025) **Author:** Daniel T. Murphy — Star-Magic UQFF Project **CVW Compliance:** v2.0.0

Abstract

This paper integrates the May 2025 ApJ study “Accessing a New Population of Supermassive Black Holes” into the Quadriadic UQFF framework. Twelve new SMBH targets ($\log M_{\bullet}/M_{\odot} = 8.7\text{--}9.7$) are resolvable at the photon-ring scale by the ngEHT (Next-Generation Event Horizon Telescope) and the proposed BHEX (Black Hole Explorer) space-based extension. The photon ring angular diameter, Eddington accretion ratio f_{Edd} , electron-to-ion temperature ratio T_e/T_i , and Blandford-Znajek (BZ) jet mechanism parameters all enter the UQFF Quadriadic layers.

1. Introduction

Following the historic imaging of M87* (2019) and Sgr A* (2022), the EHT and its successors aim to image a broader SMBH population. The photon ring angle:

$$\theta_{ring} = 2\sqrt{27} \cdot \frac{GM_{\bullet}}{c^2 D}$$

provides a direct mass-distance measurement. The May 2025 ApJ study identifies 12 new SMBH targets in the photon-ring accessible regime with ngEHT (resolution ~ 15 G λ).

2. New SMBH Targets — 12 Systems

Source	$\log M_{\bullet}/M_{\odot}$	D (Mpc)	θ_{ring} (μas)	f_{Edd}
IC 1459	9.45	29	8.9	$\sim 10^{-3}$
NGC 4594 (Sombrero)	8.82	9.8	—	$\sim 10^{-4}$
NGC 4261	8.72	31	—	$\sim 10^{-4}$
NGC 315	9.0	70	—	1.7×10^{-4}
NGC 1218	8.7	50	—	$\sim 10^{-3}$
7 additional	8.7–9.7	varied	—	$\sim 10^{-4}$ – 10^{-3}

All are Low-Luminosity AGN (LLAGN) in RIAF (Radiatively Inefficient Accretion Flow) regime.

3. Photon Ring Angular Diameter

The critical impact parameter of the photon ring for Schwarzschild BH:

$$\theta_{ring} = 2\sqrt{27} \cdot \frac{GM_{\bullet}}{c^2 D}$$

For IC 1459 ($M_{\bullet} = 2.8 \times 10^9 M_{\odot}$, $D = 29$ Mpc):

$$\theta_{ring} = 2\sqrt{27} \cdot \frac{(6.67 \times 10^{-11})(5.6 \times 10^{39})}{(9 \times 10^{16})^2 \cdot (8.96 \times 10^{23})} \approx 8.9 \mu\text{as}$$

This enters UQFF Layer 1 as:

$$g_{L1,ring} = g_{UQFF}(r, t) + \left(\frac{GM_{\bullet}}{c^2 D} \right) \cdot \theta_{ring}$$

4. Eddington Ratio and RIAF

The Eddington accretion ratio:

$$f_{Edd} = \frac{\dot{M}}{\dot{M}_{Edd}} = \frac{L_{bol}}{\eta c^2} \cdot \frac{\kappa_{es} c}{4\pi G M_{\bullet}}$$

For $f_{Edd} \ll 0.01$, the accretion is in the RIAF (hot, geometrically thick) regime. This enters Layer 2:

$$g_{L2,Edd} = f_{Edd} \cdot (T_e/T_i) \cdot \frac{GM_{\bullet}}{r^2}$$

5. Electron-to-Ion Temperature Ratio

In the RIAF, the electron and ion temperatures differ strongly:

$$\frac{T_e}{T_i} = R_{high} \cdot \frac{\beta^2}{1 + \beta^2} + R_{low} \cdot \frac{1}{1 + \beta^2}$$

where $\beta = P_{gas}/P_{mag}$, $R_{high} \approx 40$ (weakly magnetized), $R_{low} \approx 1$. For typical $\beta = 1$:

$$T_e/T_i = 40 \cdot 0.5 + 1 \cdot 0.5 = 20.5$$

6. Blandford-Znajek (BZ) Mechanism

The jet power from a spinning BH in an external magnetic field:

$$P_{BZ} = \frac{\kappa}{4\pi c} \cdot \Phi_{BH}^2 \cdot \Omega_H^2$$

where Φ_{BH} = magnetic flux threading the horizon, $\Omega_H = a_* c / (2r_H)$ = horizon angular velocity, $\kappa \approx 0.044$ (parabolic field). This enters Layer 4:

$$g_{L4,BZ} = P_{BZ} \cdot \frac{1}{r^3} \cdot a_*$$

7. Full UQFF Integration — EHT Sources

For NGC 315 ($M_\bullet = 10^9 M_\odot$, $f_{Edd} = 1.7 \times 10^{-4}$, $a_* = 0.9$):

$$\begin{aligned} g_{L1} &= g_{UQFF} + \theta_{ring} \cdot (GM_\bullet / c^2 D) \\ g_{L2} &= f_{Edd} \cdot (T_e / T_i) = 1.7 \times 10^{-4} \times 20.5 \approx 3.5 \times 10^{-3} \\ g_{L3} &= F_{U,Bi} + U_{i,buoyancy} \\ g_{L4} &= P_{BZ} / r^3 \cdot a_* \end{aligned}$$

8. ngEHT and BHEX Resolution

- **ngEHT**: $\sim 15 \text{ G}\lambda$ at 1.3 mm (EHT $\times 5$ baselines), resolves $\theta \geq$ a few μas
- **BHEX**: Space-based, planned 350+ $\text{G}\lambda$, resolves 0.1 μas class structures
- **Photon ring substructure**: BHEX can resolve the secondary ring $n = 1$ for M87*

These observational capabilities constrain a_* and $M_\bullet$ to 5%-10% precision, directly calibrating the UQFF Layer 4 BZ term.

9. Summary

Twelve new SMBH targets ($\log M/M_\text{sun} = 8.7\text{-}9.7$) are accessible to ngEHT and BHEX via photon-ring imaging. Their photon ring diameter, Eddington ratio, T_e/T_i temperature ratio, and BZ jet power all integrate directly into the Quadriadic UQFF Layers 1-4, extending the UQFF to a new population of LLAGN engines.
